

Immunotherapy Immunophenotyping

Luís Romera

This document contains the analysis of immunophenotyping of tumor tissue from mice treated with Immunocheckpoint blockade and chemoimmunotherapy.

First, let's plot the data of tumor infiltrating lymphocytes

```
#Accessing the data folder
#setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

getwd()
```

```
[1] "C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/In Vivo 2/imunofenotipagem"
```

```
#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0
4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

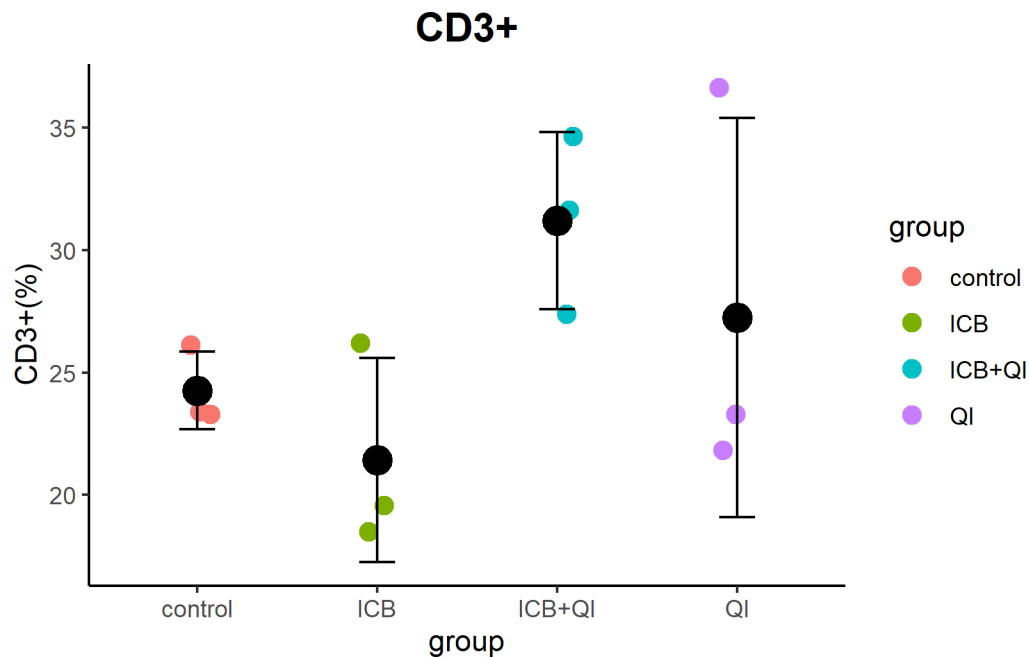
```

library(tidyverse)
library(magick)

tilPlot <- ggplot(tilData,
                  aes(
                    x = group,
                    y = CD3,
                    color = group
                  )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+",
    y = "CD3+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)

tilPlot

```



```
ggsave(
  filename = "CD3.pdf",
  plot = tilPlot,
  width = 8,
  height = 6
)
```

Now, let's plot the TCD8+ lymphocytes

```
#Accessing the data folder
#setwd("C:/Users/luizr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

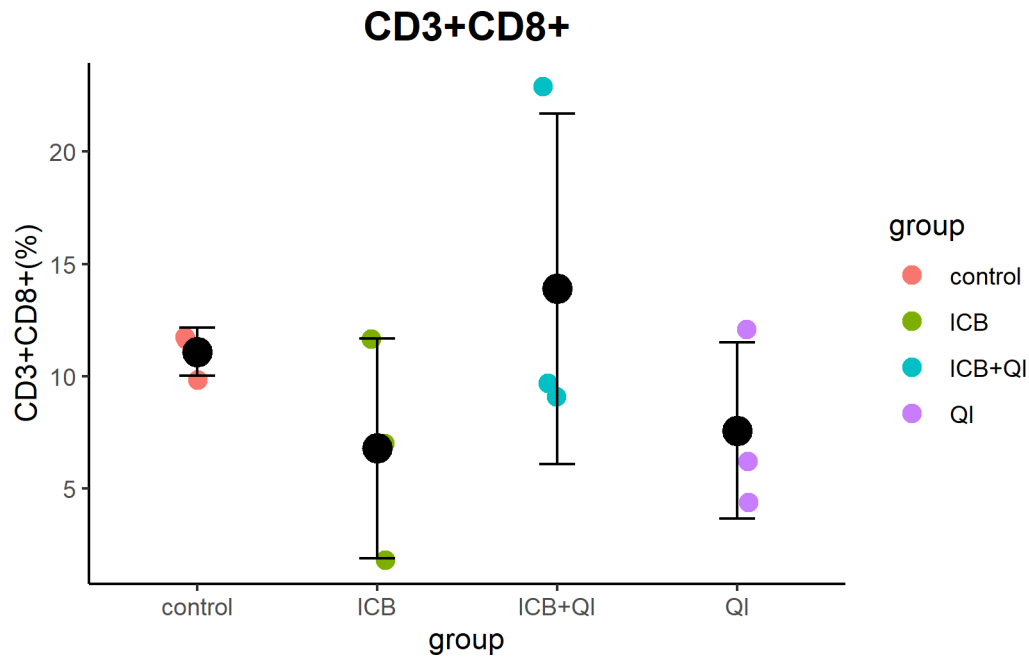
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0

4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

```
library(tidyverse)
library(magick)

til8Plot <- ggplot(tilData,
                   aes(
                     x = group,
                     y = CD3CD8,
                     color = group
                   )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD8+",
    y = "CD3+CD8+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

```
til8Plot
```



```
ggsave(
  filename = "CD8.pdf",
  plot = til8Plot,
  width = 8,
  height = 6
)
```

```
#Accessing the data folder
#setwd("C:/Users/luisr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

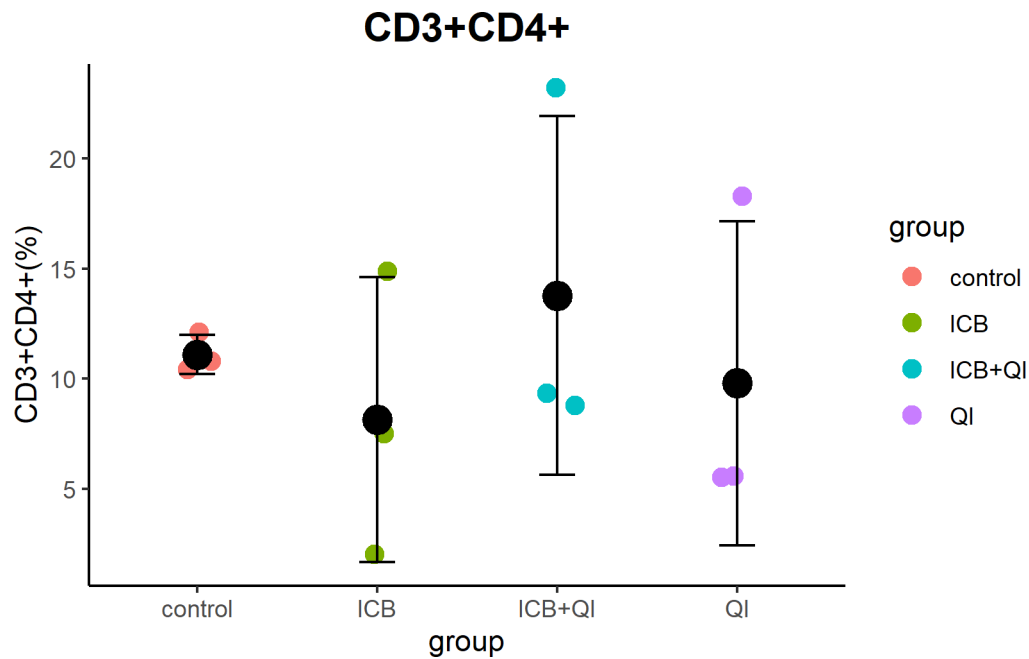
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0
4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4

6 ICB 1.82 2.02 18.5 22.2

```
library(tidyverse)
library(magick)

til4Plot <- ggplot(tilData,
                   aes(
                     x = group,
                     y = CD3CD4,
                     color = group
                   )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD4+",
    y = "CD3+CD4+(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

til4Plot



```
ggsave(
  filename = "CD4.pdf",
  plot = til4Plot,
  width = 8,
  height = 6
)
```

Now, ,let's plot exhaust TCD8+ lymphocyte

```
#Accessing the data folder
#setwd("C:/Users/luizr/Documentos/Documents/Documentos - Copia/Mestrado/imunofenotipagem")

#Importing data
tilData <- read.csv("tils.csv")

head(tilData)
```

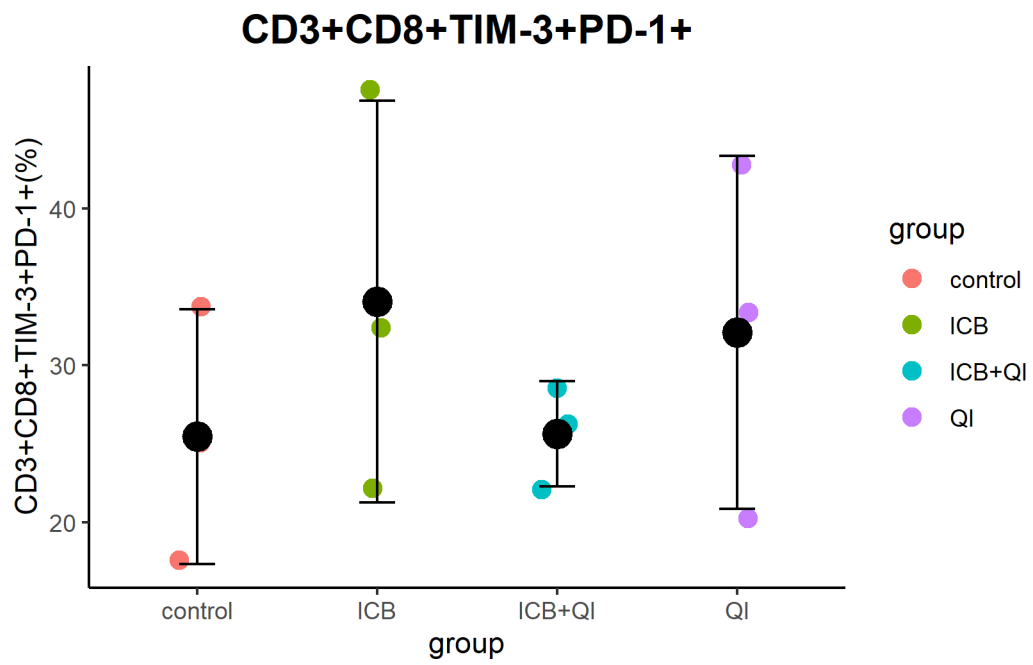
	group	CD3CD8	CD3CD4	CD3	exhaust
1	ICB+QI	22.90	23.20	34.6	28.6
2	ICB+QI	9.66	8.81	31.6	26.3
3	ICB+QI	9.13	9.35	27.4	22.0

4	ICB	11.60	14.90	26.2	47.6
5	ICB	6.95	7.52	19.6	32.4
6	ICB	1.82	2.02	18.5	22.2

```
library(tidyverse)
library(magick)

tilexPlot <- ggplot(tilData,
                    aes(
                      x = group,
                      y = exhaust,
                      color = group
                    )) +
  geom_jitter(
    width = 0.1,
    size = 3
  ) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 5,
    color = "black"
  ) +
  stat_summary(
    fun.data = mean_sdl,
    fun.args = list(mult = 1),
    geom = "errorbar",
    width = 0.2,
    color = "black"
  ) +
  theme_classic() +
  labs(
    title = "CD3+CD8+TIM-3+PD-1+",
    y = "CD3+CD8+TIM-3+PD-1(%)"
  ) +
  theme(plot.title = element_text(
    hjust = 0.5,
    face = "bold",
    size = 15
  ))
)
```

tilexPlot



```
ggsave(  
  filename = "exhaust.pdf",  
  plot = tilexPlot,  
  width = 8,  
  height = 6  
)
```